

Performance Testing

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust, Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	ML Optimization Core Engine (/api/v1/predict-yield endpoint)
Test Environment	Cloud / Hosted Server (AWS EC2 t3.medium instance staging environment)
Test Date	12 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline simulation benchmarking standard dashboard traffic requests to the Telemetry API.	50	300	Response latency remains under 200ms with 0% error packet drops.
2	Peak harvest-season data sync load concurrent hits targeting the Crop Yield Prediction Engine.	200	600	Average response time stays below 500ms; HTTP 200 OK success codes returned.
3	Stress testing rapid micro-climate sensor array batch submissions to check data pipeline queuing limits.	500	120	System processes requests gracefully via message queue without dropping data entries.
4	Sudden weather alert notification spike stress test simulation hitting regional telemetry broadcast workers.	1000	60	Server utilization spikes up to 85% but successfully self-throttles without core service disruption.